

Relations

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Problem 1. Let G be a group and S a subset of G . Show that there exists a unique smallest subgroup of G that contains S . Describe it explicitly, in terms of "words". Show it is the intersection of all subgroups that contains S , and show it is contained in every subgroups that contains S .

Proof. Let $G = \langle S \rangle$ thus there exists subgroups of G containing S . Suppose $S \subset H$ where H is the smallest subgroup of G containing S . Then $\forall a \in \langle S \rangle$ by definition $a = s_1^{\pm 1} s_2^{\pm 1} \dots s_k^{\pm 1}$ where $s_i \in S$ and i denotes the position in a . But since H is a group containing S and is closed under multiplication $s_1^{\pm 1} s_2^{\pm 1} \dots s_k^{\pm 1} \in H$ thus $a \in H$. Thus $\langle S \rangle \subseteq H$. However since H is the smallest subgroup of G containing S if $\langle S \rangle \subseteq H$ then $\langle G \rangle = H$. $\langle S \rangle = \{\prod_{i=1}^k s_i^{\pm 1}, e \mid s_i \in S, k \in \mathbb{N}\}$. $\langle S \rangle = F(S \cup S^{-1} \cup \{e\})$ since $\forall s \in \langle S \rangle$ $s \in F(S)$ and $\forall s \in F(S)$ $s \in \langle S \rangle$.

Suppose $S \subset H$ where H is subgroup of G containing S . Then $\forall a \in \langle S \rangle$ by definition $a = s_1^{\pm 1} s_2^{\pm 1} \dots s_k^{\pm 1}$ where $s_i \in S$ and i denotes the position in a . But since H is a group containing S and is closed under multiplication $s_1^{\pm 1} s_2^{\pm 1} \dots s_k^{\pm 1} \in H$ thus $a \in H$. Thus $\langle S \rangle \subseteq H$.

Because $\langle S \rangle \subseteq H$ for any subgroup H containing S . Then by definition of intersections, $\langle S \rangle \subseteq \bigcap_{S \subset H \subset G} H$. Furthermore, since $\langle S \rangle$ is one such H containing S and it is the smallest thus there $\nexists H$ s.t $\langle S \rangle \not\subseteq H$. Then $\bigcap_{S \subset H \subset G} (H) = (\bigcap_{S \subset H \neq \langle S \rangle \subset G} (H)) \cap \langle S \rangle \subseteq \langle S \rangle$. Thus $\langle S \rangle = \bigcap_{S \subset H \subset G} (H)$. $\square$

Problem 2. Do the "same" thing for normal subgroups – ie show there is a unique smallest normal subgroup that contains S , and describe it both explicitly, and as an intersection.

Proof. Let N_0 be the smallest normal subgroup which contains S . Then $N_0 = \{\prod_{i=1}^k g_i s_i^{\pm 1} g_i^{-1}, e \mid s_i \in S, g_i \in G, k \in \mathbb{N}\}$.

Let N_1 and N_2 be normal subgroups of G . Then let $a \in N_1 \cap N_2$. Since N_1 and N_2 are normal $gag^{-1} \in N_1$ and $gag^{-1} \in N_2 \forall g \in G$. Thus $\forall g \in G$, $gag^{-1} \in N_1 \cap N_2$. Thus $N_1 \cap N_2 \triangleleft G$.

Thus $\bigcap_{S \subset N \triangleleft G} N$ is a normal subgroup. $S \subset \bigcap_{S \subset N \triangleleft G} N$. Since N_0 is one such N , $\bigcap_{S \subset N \triangleleft G} N = N_0 \cap \bigcap_{S \subset N \triangleleft G} N \subseteq N_0$. But since N_0 is the smallest normal subgroup of G containing S , then $\bigcap_{S \subset N \triangleleft G} N$ cannot be smaller than it, thus $\bigcap_{S \subset N \triangleleft G} N = N_0$ $\square$

Problem 3. Let S be a set, and R a set of words in S . Let $F(S)$ be free on S , and let $N(R)$ be the smallest normal subgroup that contains all the words R . Let $q : F(S) \rightarrow Q$ be the quotient group. Note we have a natural $s : S \rightarrow Q$ (that might not be injective). Thus each word in S gives us an element of Q . Show that every word in R is trivial in Q , and that $S \rightarrow Q$ is universal with this property: Namely, if G is any other group, and we have a map $t : S \rightarrow G$ (a map of sets, ie a function) then $t(w)$ is trivial for all w in R iff t factors through s (ie $t = T \circ s$ for some group homo $T : Q \rightarrow G$) and in that case the factoring map $T : Q \rightarrow G$ is unique.

Proof. Let S be a set, and R a set of words in S . Let $F(S)$ be free on S , and let $N(R)$ be the smallest normal subgroup that contains all the words R . Let $q : F(S) \rightarrow Q$ be the quotient group $Q := F(S)/N(S)$. Note we have a natural $s : S \rightarrow Q$. $R \subset N(R)$, then by definition of the quotient map, $q(R) \subset N(R) = eN(r)$. Thus R is trivial in Q .

Since $F(S)$ is a free group on S . Let $s : G \rightarrow F(s)$. Then for any group G and for any map $t : S \rightarrow G$ there exists a unique homomorphism $g_t : F(S) \rightarrow G$ s.t $t = g \circ s$. Restrict t to the maps such that the kernel of unique homomorphism g_t contains R . Because the $R = \ker q \subseteq \ker g_t$, by the prehomework theorem about groups $\exists p$ s.t $g_t = p \circ q$. Since q is a homomorphism and g_t is a homomorphism p must be a homomorphism. Furthermore since g_t is unique and q is surjective then p must also be unique. Thus for any map t s.t $\ker g_t = R$ there exists a unique homomorphism from $g'_t : Q \rightarrow G$ s.t $t = g'_t \circ q \circ s$. Thus Q is universal whenever $N(R)$ is trivial under any map to a group. $\square$

Problem 4. Show that if $s_i : S \rightarrow Q_i$, $i = 1, 2$ each have this universal property (for groups Q_i), then the Q_i are canonically isomorphic: show there exists unique homos $f_{ij} : Q_i \rightarrow Q_j$ such that $f_{ij} \circ s_i = s_j$, and these are inverse isomorphisms. So this shows that (up to canonical isomorphism) there is only one group with this property, we write $Q =: \langle S \mid R \rangle$ "the group generated by S , with the relations R ".

Proof. Let Q_i $i = 1, 2$ each have the universal property seen above in question 3. Then R is trivial in Q_1 and Q_2 . Then by definition there exists a unique homomorphism from $g_t : Q_i \rightarrow G$ (G is any group) s.t $t : S \rightarrow G$ factors through s_i and $R \subset \ker g_t$. Since R is trivial in Q_1 and Q_2 for any map $s_2 : S \rightarrow Q_2$ the kernel of the homomorphism $f_{1,2} : Q_1 \rightarrow Q_2$ contains R and the homomorphism is unique and $f_{1,2} \circ s_1 = s_2$ by the definition of the universal property. Similarly for any map s_1 the kernel of the homomorphism $f_{2,1} : Q_2 \rightarrow Q_1$ contains R and thus the homomorphism is unique and $f_{2,1} \circ s_2 = s_1$ by definition of the universal property. Since $f_{1,2}$ and $f_{2,1}$ are unique homomorphism, then $g_1 = f_{1,2} \circ f_{2,1}$ must also be unique homomorphism from $Q_1 \rightarrow Q_1$ by definition of function composition. The identity is by definition a homomorphism from $Q_1 \rightarrow Q_1$ and $s_1 = id_{Q_1} \circ s_1$ and thus is the unique homomorphism under s_1 . And since g_1 is also a unique homomorphism under s_1 , then $g_1 = id_{Q_1} = f_{1,2} \circ f_{2,1}$. Since $f_{1,2}$ and $f_{2,1}$ are unique homomorphism, then $g_2 = f_{2,1} \circ f_{1,2}$ must also be unique homomorphism from $Q_2 \rightarrow Q_2$ by definition of function composition. The identity is by definition a homomorphism from $Q_2 \rightarrow Q_2$ and $s_2 = id_{Q_2} \circ s_2$ and thus is the unique homomorphism under s_2 . And since g_2 is also a unique homomorphism under s_2 , then $g_2 = id_{Q_2} = f_{2,1} \circ f_{1,2}$. Similarly with symmetric logic, $id_{Q_2} = f_{2,1} \circ f_{1,2}$. Thus $f_{1,2}$ and $f_{2,1}$ are left and right inverses of each other and thus are bijections and homomorphisms and thus are inverse isomorphisms of each other. $\square$

Problem 5. Describe $G = \langle X_1, \dots, X_n \mid X_i X_j X_i^{-1} X_j^{-1}, i, j = 1, \dots, n \rangle$ - identify this (with proof) with a group we know.

Proof. By the relation, $X_i X_j X_i^{-1} X_j^{-1} = e$ for $\forall i, j \in \mathbb{N}$. Thus $X_i X_j X_i^{-1} = X_j$ and $X_i X_j = X_j X_i$. Thus G is a free abelian group and is $\mathbb{Z}^n$. $\square$

Problem 6. Find the order of $G = \langle I, J \mid I^2 = J^2, J^4 = 1, IJI^{-1} = J^{-1} \rangle$, and identify this with a concrete group (that we may not yet have discussed).

Proof. By the relations we know, $I^2 = J^2$, $J^4 = e$, and $IJ = J^{-1}I$. Thus for all $a \in \mathbb{Z}$, $I^a J = J^{-a} I^a$. Thus the terms can be moved around by following the relations and negating the exponent of J every time an I swaps its position with it. Thus $\forall g \in G$, $g = I^a J^b$ and since $I^2 = J^2$ and $J^4 = 1$ $a = 0, 1$ and $b = 0, 1, 2, 3$. Thus there are at least 8 possibilities for g. $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ and the relations on the quaternion group are as follows $i^2 = j^2 = k^2 = -1$ and $j^4 = 1$ so $i^4 = 1$ Furthermore $ijk = -1$ and $ijk^2 = -ij = -k$ so $ij = k$. Thus $ijk = ijij = -1$. Thus $ijij^2 = -j$, $-iji = j$ and $-iji^2 = -ij(-1) = ij = ji$. Thus the quaternion group follows all the relations of G and has order 8 thus G must have order 8. There is a injective homomorphism from the quaternion group to the G. Thus they are isomorphic. $\square$